

Applications of fiber bundles in gauge field theory

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Fiber bundle

Lie algebra on G (especially $SU(3)$)

exponential mapping: $\text{Exp}: V_e \rightarrow G$, $\exp^{-1}(g^\gamma) = \gamma T_g^a \in V_e$

For Lie algebra of $SU(n)$ ($\dim SU(n) = n^2 - 1$),

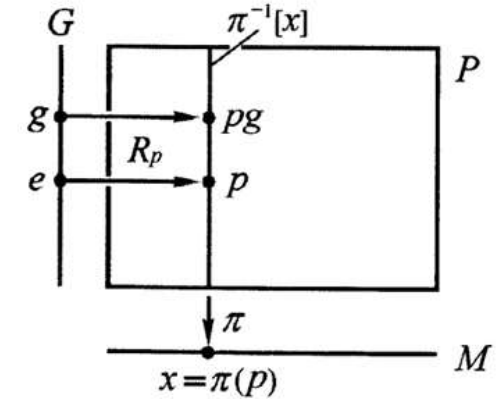
We represent $g = \exp(\gamma T_g^a) = \exp\left(i \frac{\theta_a \lambda_a}{2}\right) = e^{i\theta_a T^a}$

$$SU(n) = \{U \in U(n) | \det U = 1\}$$

$$\text{Lie Algebra } SU(n) = \{A \in U(n), \text{tr} A = 0\}$$

$$A^\dagger = -A, \{m \times m \text{ antihermit matrices}\}$$

Lie bracket: $[T^a, T^b] = i f^{abc} T_c$, f^{abc} is structure constant



$$\begin{aligned} \lambda^1 &= \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & \lambda^2 &= \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & \lambda^3 &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \\ \lambda^4 &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, & \lambda^5 &= \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, & \lambda^6 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \\ \lambda^7 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, & \lambda^8 &= \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}, \end{aligned}$$

When $m=3$, A can be represented as these λ

Fiber bundle

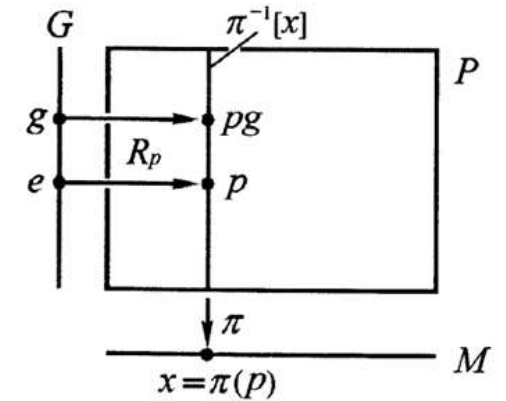
Principle Fiber bundle (P, M, G)

P : bundle manifold, $\dim m + n$

M : base manifold, $\dim m$

G : structure group (Lie group), $\dim n$, right acts $R : P \times G \rightarrow P$

$\pi : P \rightarrow M$, projective mapping, reversible, $\pi^{-1}[x]$ called fiber
 $\sigma(x) : U \text{ in } M \rightarrow P$, section (if $U = M$, it is global)



Right acts: $R : P \times G \rightarrow P$, can be written as $R_g : P \rightarrow P$, or $R(p) : G \rightarrow P$

It doesn't influence $x = \pi(p)$

Fiber bundle

fiber bundle associated to $P(M,G) : (Q,M,F)$

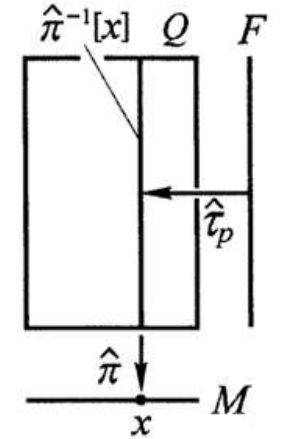
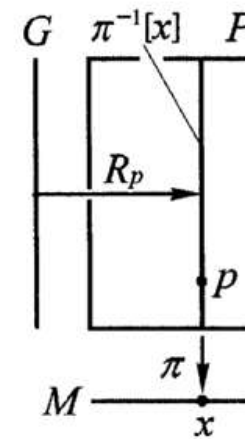
$$Q: q \in Q, q = p \cdot f = pg \cdot g^{-1}f$$

M : base manifold ,dim m

F : type fiber , a manifold left-invariant by g in G

$\hat{\pi}: Q \rightarrow M$, projective mapping ,reversible

$\hat{\sigma}: U \text{ in } M \rightarrow P$, section (if $U = M$, it is global)



Fiber bundle

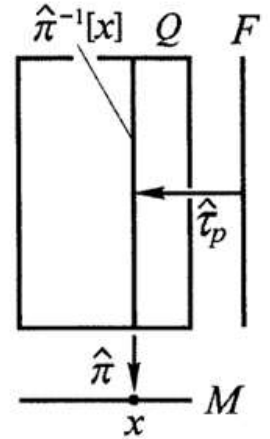
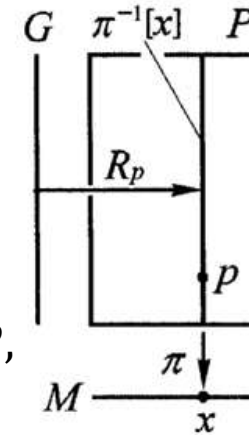
Tangent space T_p :

Vertical subspace $V_p := \{ X \in T_p P \mid \pi_*(X) = 0 \}$

Horizontal subspace $H_p \subset T_p P$,

$$(b) R_g * [H_p] = H_{pg}, \forall p \in P, g \in G, (a) T_p P = V_p \oplus H_p, \quad \forall p \in P,$$

(c) H_p of p are smooth



Connection in Principle fiber bundle :3 equivalent definition:

Horizontal space

a 1-form field: $\tilde{\omega} \in \text{Lie Algebra of } G$

Later will show in part 3

$$\tilde{\omega} \in \mathcal{G}, \quad \omega_p \text{ is } C^\infty$$

$$(a) \tilde{\omega}_p(A_p^*) = A, \quad \forall A \in \mathcal{G}, p \in P,$$

$$(b) \tilde{\omega}_{pg}(R_{g*}X) = \text{Ad}_{g^{-1}}\tilde{\omega}_p(X), \quad \forall p \in P, g \in G, X \in T_p P.$$

Yang-Mills gauge field(QCD)

quark's quantum field : spinor field with 3 color
it is a inner space with transition U in SU(3)

It's Lagrangian density :

If θ is not relevant with x: $A_\mu = 0$ is correct(global gauge invariance)
Local gauge invariance: $\theta(x)$: introduce field translation $A_\mu \rightarrow A_\mu'$

$$\psi = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \end{pmatrix},$$

$$\mathcal{L}_q = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi,$$

$$\mathcal{L}_q = \bar{\psi}(i\gamma^\mu D_\mu - m)\psi, \quad D_\mu = \partial_\mu + iqA_\mu,$$

Yang-Mills gauge field (QCD)

A_μ is gauge potential. (gauge field of SU(3), electromagnetic field of U(1))

In order to keep the Invariance of the Lagrangian:

$$\psi' = U\psi = e^{i\theta^a T_a} \psi$$

$$A_\mu = A_\mu^a T_a ,$$

$$A_\mu^{a'} = A_\mu^a - f^{abc} \theta^b A_\mu^c - \frac{1}{g} \partial_\mu \theta^a = A_\mu^a - \frac{1}{g} D_\mu^{ab} \theta^b$$

Description of gauge field in fiber bundle

Into fiber bundle theory :

Fiber bundle : base manifold $M = \text{Spacetime Manifold}$
gauge group $G = \text{SU}(3)$
typical fiber $F = V_G = \mathcal{C}^3$, represent space

let $\rho(g): G \rightarrow \hat{G}, \rho(g) = \exp^{-1}(g), \hat{G}$ is the representation group of G

transition between cross sections:

$$\begin{aligned}\sigma' \rightarrow \sigma : \quad \sigma'(x) &= \sigma(x)g^{-1}(x) , , \\ \phi(x) &\in \hat{G} , \quad \phi(x)' = \rho(g(x))\phi(x) , \\ f'(x) &= \rho(g(x))f(x)\end{aligned}$$

Description of gauge field in fiber bundle

There is an invariance :

$$\Phi(x) = \sigma(x) \cdot f(x), \quad \Phi(x) = p \cdot f \in Q ,$$

$\Phi(x)$ is a section of Q

In tangent bundle :

$$v(x) = \left(x, e_\mu(x) \right) \cdot f^\mu = e_\mu(x) f^\mu(x)$$

is covariant.

So we use $\Phi(x)$ to represent quack field , $\sigma(x)$ represent gauge choice

Description of gauge field in fiber bundle

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How about connection?

Connection definition 3:

$\forall$ local trivial $T_U: \pi^{-1}[U] \rightarrow U \times G$ and $T_V: \pi^{-1}[V] \rightarrow V \times G$, $(U \cap V \neq \emptyset)$, with $g_{UV}: T_u \rightarrow T_v$
Given a 1-form field $\omega_U \in \text{Lie Algebra of } G$, fit:

$$\omega_V(Y) = \text{Ad}_{g_{UV}(x)}^{-1} \omega_U(Y) + L_{g_{UV}(x)}^{-1*} g_{UV*}(Y), \quad \forall x \in U \cap V, \quad Y \in \mathcal{T}_x M$$

Description of gauge field in fiber bundle

For gauge potential :

$$A_\mu^{a'} = A_\mu^a - \frac{1}{q} D_\mu^{ab} \theta^b$$

Where $\mathbf{A} = A_\mu^a dx^\mu e^a$, can be correctly fit definition 3:

So we knew that $k\mathbf{A}$ is the connection ω

Further more ,the 2-form $\mathbf{F}$ is correctly the

Cartan's second structure equation:

$$R_\mu^\nu = d\omega_\mu^\nu + \omega_\mu^\lambda \wedge \omega_\lambda^\nu$$

can be related with $\mathbf{F} = d\mathbf{A} + \mathbf{A}^2$ (which will be proved later)

So we knew : $\mathbf{F}$ can be the curvature of fiber bundle

Differential form of gauge field and gluon

Proof of $\mathbf{F} = d\mathbf{A} + \mathbf{A}^2$

$$\mathbf{A} = A_\mu^a dx^\mu e^a, \text{ gauge transform: } U, UU^+ = I,$$

$$\mathbf{A}' = U\mathbf{A}U^+ + UdU^+, (dU^+ = \partial_\mu U^+ dx^\mu, 1\text{-form}),$$

$$d\mathbf{A}' = Ud\mathbf{A}U^+ + dU\mathbf{A}U^+ - U\mathbf{A}dU^+ + dUdU^+,$$

$$\mathbf{A}'^2 = U\mathbf{A}^2U^+ + U\mathbf{A}dU^+ + UdU^+U\mathbf{A}U^+ + UdU^+UdU^+,$$

Since the Lagrangian of the field needs to be integrated over the entire space, total differentials can be added to the equation without affecting the result.

$$\mathbf{A}'^2 = U\mathbf{A}^2U^+ + U\mathbf{A}dU^+ - dU\mathbf{A}U^+ - dUdU^+,$$

$$\mathbf{F} = \mathbf{A}^2 + d\mathbf{A}, \text{ where } \mathbf{F}' = \mathbf{A}'^2 + d\mathbf{A}' = U(d\mathbf{A} + \mathbf{A}^2)U^+, \text{ covariance}$$

When described in component form, the result is:

$$F_{\mu\nu} = [A_\mu, A_\nu] + \partial_\mu A_\nu - \partial_\nu A_\mu$$

Differential form of gauge field and gluon

When we write the Lagrangian of gauge fields, we obtain two terms: the self - energy term and the term interacting with quarks.

According to the QCD Feynman diagram rules, the gauge field's carrier particles are named as gluon:

(today we don't have to introduce the ghost field)

$$L_A = -\frac{1}{2} F_{\mu\nu}^a F_b^{\mu\nu} \text{tr}(T^a T^b) ,$$

$$L_I = -g \bar{\psi} \gamma^\mu A_\mu \psi ,$$

